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A Short Editorial:

YSJOI - by the youth, for the youth of Iraq!

In the great land that was once Mesopotamia, and the home of the hanging gardens of Babylon, progress in science has slowed.

Why?

Because the youth of Iraq have no platform to build upon! Opportunities to learn about science, technology, engineering, and mathematics are rare, and students trying to learn about them are even actively discouraged – I speak from experience.

But instead of surrendering to that silence, I have chosen to rally the youth of Iraq. If we do not have a base to stand on, we will build one!

And the first step to that is this Youth Science Journal Of Iraq, YSJOI, where we aim to provide that very base – an opportunity to teach, learn and explore through different articles – be it inspiring them, writing them, or reading them!

Thank you for being part of this beginning.
Now, dive in – into the weird, the beautiful, and the wonderful.

– Satvik M. Bhure, Founder of YSJOI

Question I: How do we know that infinite mathematical series do converge?

– Mahmood Ardalan, 15

How do we know that infinite series do converge?

Infinity at its core is a very complicated number. In fact, some would even say that it is not a number at all. So how can we find the 'infinite' sum of anything, especially a series? Should the infinite sum of anything not just be infinity?

While counterintuitive, an infinite series can have a finite sum. But the question is: how can we prove that? Formally, because we cannot add infinite sums; which is theoretically, mathematically, and practically impossible, we can instead find convergence as a sequence of partial sums.

The convergence of the series can only exist if the sequence of those sums approaches a finite number as the sequence goes to infinity. More simply, there must be a maximum sum to which the terms in the sequence add up. If the limit exists and is not infinite, we determine that the series converges. If the limit goes to infinity or bounces around without settling on a number, we say that the series diverges; there is no sum to infinity. Essentially, there must come integer values of positions in the sequence where the values added by consecutive terms shrink so quickly and are so insignificantly small that they stop making noticeable differences in the total. Since it is rare for us to be able to find a regular algebraic formula which works for the infinite sum, we can instead test this limit: for example, if comparing series between zero, the question series, and a larger series than the question series, we can see that if the larger series converges, then the smaller series must always converge as well.

There is the infinite test used in geometric sequences as well if the consecutive ratio between terms is less than one, then the series will shrink and converge.

We know that infinite series converge because their sum is separate from the whimsy of regular infinity; instead, they just converge to a limit that is similar to their sum to infinity, which it meets when their number becomes infinity.

Question II: Could neutron stars contain a state of matter that cannot exist anywhere on Earth?

– Shirin Kawan, 14

Imagine squeezing a star more massive than the Sun into a space roughly the size of a city. The result would be a neutron star, one of the strangest objects in the universe. These stars are formed when massive stars reach the end of their lives and collapse under their own gravity. What makes them particularly fascinating is not just their extreme density, but the possibility that matter inside them exists in a form that we cannot reproduce on Earth.

Neutron stars are incredibly dense. A small amount of their material would have an enormous mass compared with anything we encounter in everyday life. Under such extreme pressure, atoms cannot maintain their usual structure. Electrons combine with protons to form neutrons, leaving behind matter made mostly of tightly packed neutrons. This is very different from ordinary matter, which

consists of atoms with nuclei surrounded by electrons. However, scientists believe that the matter inside a neutron star may be even more unusual than this. Deep within its core, the pressure and density could become so extreme that neutrons themselves break down into their smaller components, called quarks. Another possibility is that matter enters an exotic state involving particles that are rarely found under normal conditions.

Although these ideas are supported by theoretical physics, scientists have not yet confirmed exactly what exists at the centre of a neutron star. Could such matter ever exist on Earth? In laboratories, scientists can create extremely high temperatures and briefly produce unusual forms of matter, such as quark-gluon plasma. However, the conditions inside a neutron star are very different. Its immense gravitational pressure cannot be reproduced in a laboratory, and any exotic matter created under experimental conditions may exist only for an extremely short time.

Ultimately, neutron stars may contain forms of matter that cannot exist naturally or remain stable on Earth. While much remains unknown, these extraordinary objects remind us that the universe can create conditions far beyond our current technological abilities. What lies at the centre of a neutron star may eventually change our understanding of matter itself.

Question III: Is empty space truly empty?

– *Ahmed Bilal, 16*

Modern physics is unanimous on this topic: space is not empty. It is entirely full of energy. While we traditionally define a vacuum to mean "nothing," achieving absolute nothingness on a subatomic level is physically impossible. Even if you emptied out every single atom from a specific region of space and kept it completely protected from all external radiation, the zone would still not be devoid of activity. It would inherently possess a foundational ground energy.

Vacuum energy exists primarily because of Quantum Field Theory, which posits that the fundamental nature of reality is made up of fields rather than particles. In this framework, particles are simply localized excitations of these underlying fields. You cannot extract a field from the universe; you can empty out all the physical particles to create a technical vacuum, but the field itself remains. Furthermore, you cannot prevent a field from vibrating. Because of Heisenberg's Uncertainty Principle, every field has a lowest limit of vibration it can exist at, which is called its zero-point energy. These underlying fields constantly undergo rapid fluctuations where particles of energy spontaneously appear and disappear. We call these transient phenomena virtual particles. Virtual particles have real, measurable consequences in our physical world. For example, the energy of the quantum vacuum can generate a physical force that causes closely spaced metal plates to move together. This distinct phenomenon is known as the Casimir effect. Beyond these subatomic field fluctuations, the entire universe is also continually bathed with the Cosmic Microwave Background radiation and dark energy. Ultimately, you cannot have a true, empty vacuum because a vacuum is not the absence of reality; rather, it is an active part of reality itself. Space will always buzz with quantum activity, proving that empty space is never truly empty.

Question IV: Did the ingredients of life originate on Earth, or did chemistry delivered by meteorites play a crucial role?

– Mustafa Barzani, 14

Well, both played a crucial and massive role in the beginning of life. But who's the star of the show?

Earth was obviously a major participant in this process of beginning life, and unbeknownst to most ordinary people, Earth had synthesized many simple chemicals into complex organic compounds. Furthermore, it housed minerals like clay and tourmaline, which act as reactive surfaces that can help make early RNA-like polymer chains.

It seems like Earth didn't really need meteorites to start life, as it already had so many of the building blocks it required, but that's not the full story, as meteorites played a very major role in early life, and their contributions can't be ignored.

Most people see meteorites as flying rocks that crush and ruin the Earth's surface. Though looking at the bigger picture, we realize that these deliveries from outer space actually provide many of the major elements needed for life. These floating rocks carried thousands of organic molecules onto Earth, including amino acids, DNA and RNA nucleobases, and sugars. The story doesn't stop there though, as meteorites also supplied earth with crucial compounds and elements, like water and zinc, with studies showing a major portion of these compounds and elements can form bodies in the outer-solar-system. Furthermore, they also delivered multiple phosphorus compounds, which are the literal backbone of genetic material.

Neither the isolated planet, nor the giant asteroids from outer space can provide the full set for beginning life as both are missing blocks the other has. Earth provided the perfect incubator, and the meteorites planted the seeds.

So not all ingredients of life originated from Earth, some did, and life couldn't have begun without the help of meteorites and their compounds.

Question V: If we can rewrite DNA, why can't we reliably predict the organism we will create?

– Zanyar Ahmadi, 13

If we can rewrite DNA, why can't we reliably predict the organism that we will create?

Scientists have figured out how to edit genes to their liking with tools like CRISPR, but why can't they predict exactly what the result of editing a particular gene will be?

DNA is usually rewritten to alter some characteristic of an organism, but most of the time this characteristic is controlled by tens of thousands of genes that each play a small role in this characteristic and changing one gene may also alter another gene which may cause an unwanted side effect or cause the gene edit to become ineffective. Moreover, a single gene often controls multiple, seemingly unrelated traits which leads to further side effects that could be harmful. Just the height of a human has more than 12,000 different genes controlling it!

Besides just the DNA structure, many other external factors also affect the behavior of a characteristic. Environmental factors such as nutrition, temperature, climate, stress, etc. all play a role in how much a characteristic is developed. Two people may have the exact same DNA sequence but raised in different places will each develop differently due to surrounding factors and some random cellular events that could occur during growth.

Finally, the presence of a gene does not guarantee it will actually do anything, due to a process called gene expression. Cells produce compounds called chemical tags, methyls and acetyls, which latch onto a gene and lock it down or opens them up to be expressed based on signals from the cell, so if an edited gene gets locked it becomes basically useless, this just makes the entire process much more difficult and uncertain.

Now the reasons why DNA editing isn’t fully predictable are clear, the process is a lot more complicated than just changing a genes code to match a property a person seeks, there are many underlying things that have to be taken into consideration as well as the risks.

Question VI: Why are elements able to bond with one another?

– Ibrahim Yousif, 14

Have you ever wondered how tiny atoms come together to form the world around us? From the water we drink to the air we breathe, almost everything we see depends on atoms joining together.

But if atoms are so tiny, what makes them connect in the first place? How are they brought together? Most of us have learned from school in the 8th grade that the main reason atoms bond is to become more stable. An atom contains electrons arranged in energy levels around its nucleus, and the electrons in its outermost shell are especially important. Atoms are generally more stable when their outer electron shell is full. Some atoms achieve this by losing electrons, others by gaining electrons, while some share electrons with other atoms.

By doing so, they can reach a more stable electronic arrangement, often similar to that of a noble gas. In simple terms, atoms bond because the result can be more energetically favorable than remaining separate. There are different types of chemical bonds, each with its own way of bringing atoms together.

Ionic bonding happens when electrons are transferred from one atom to another, forming positively and negatively charged ions that attract each other, also known as cations and anions. For example, sodium and chlorine form sodium chloride, which is your ordinary table salt!

Covalent bonding, on the other hand, involves atoms sharing pairs of electrons. Water is a familiar example, as hydrogen and oxygen atoms are held together by covalent bonds.

Metals form another type of bond called metallic bonding, in which positive metal ions are attracted to delocalised electrons that can move throughout the structure, the delocalized electrons are also the reason metals can conduct electricity! These different bonding methods help explain why substances have such different properties, from the softness of some metals to the strength of diamond.

Ultimately, chemical bonding is one of the most important ideas in chemistry because it explains how simple atoms combine to create an incredible variety of substances. The food we eat, the medicines that help us recover, and even the materials used to build our homes - they all depend on atoms interacting in different ways!

Question VII: Can we erase or rewrite unwanted memories?

– Soran Danyal, 14

Our memories are an integral part of who we are. They remind us of the happy moments with the people we love, the joyful times of our childhood, and the nostalgic feelings of our old houses. But they can also hold experiences we wouldn't want to relive, things we've done or things we've seen that we'd want to take back. Could we, by any chance, alter those memories?

Medicine is advancing, and neuroscience is leaping with every new discovery. Further research is being done, and several interesting things have already been found. For example, whenever you recall a memory, good or bad, you are in the cycle of reconsolidation, which has three steps:

1. Reactivation: This is when you go through the experience again, living it once more, getting all the feelings and senses associated with it.
2. Labile phase: This is a period of time, for about 2-6 hours, when the neurons holding the memory physically loosen, leaving the memory vulnerable to disruption.
3. Stabilization: This is when new proteins are synthesized to help put those neurons back in shape, solidifying the memory.

Neuroscientists and psychiatrists realized that the labile phase is the time to strike – and developed several different methods to help people, such as PTSD patients, whose memories are working against them.

One way that is commonly used is by means of medicine. The patient's vitals are observed, and it is made sure that the medicine used, usually propranolol, is safe to administer.

Once sure, the doctors give the patient a dose of the medicine, and wait until it has spread throughout the bloodstream. They then proceed to ask the patient to write down a detailed account of the traumatic experience, going through the feelings in detail. In the following sessions, instead of writing the experience, the patient simply reads their own script. The propranolol is a beta-blocker, which helps reduce the intensity of the feelings during the reactivation phase, making the memory induce less stress, even after stabilization. After several times, the feelings are barely there.

So no, we are unable to erase or change the memories themselves, but we can alter the impact they have on us, giving them less control.

A CELEBRATION OF YSJOI'S FIRST YEAR

The Youth Science Journal Of Iraq. YSJOI. An endeavor that is celebrating its first year this month.

Only a year ago, YSJOI was merely an idea. An idea that might not have worked. An idea so audacious that I might just pull it off.

Today, we have 13 issues, 77 articles, and a team that is ready every month to find and answer more and more questions for the youth.

I can only express my gratitude to everyone that has helped me develop YSJOI, and say wholeheartedly that we will do our level best to keep building YSJOI as long as we can. We have more ideas planned for this year, to bring science to the youth of Iraq in more and more ways, to answer more questions, and to defeat any barriers that stand against access to STEM.

And we hope that you will all continue to support YSJOI as it works to build the opportunities it could not find before.

That is why this issue detracts from our norm of 5 articles per issue. We wanted to show that we were ready to go above, and beyond, for whatever needs to be done to help further the mission of YSJOI.

Acknowledgements

I (Satvik M. Bhure) have many people to thank for this incredible piece of work!

We thank all the scientists who have worked thus far to find all about the world so we can learn about it, and everyone who has supported any member of the YSJOI team in furthering our mission.

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And we thank you, reader, for supporting our
mission to help the Iraqi youth where there
has been no help as yet!

CONCLUSION

We hope to further YSJOI's mission to empower the youth of Iraq in STEM fields by our monthly magazine, and sincerely hope we were able to help you and other students with this monthly issue! We are always open to feedback and questions, and everyone is of course welcome to learn in their own way; after all, we are all different, and we all belong.

Make sure to visit our website at https://YSJOI.is-great.org

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